

# SHIVAM OJHA

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## SUMMARY

Research Engineer specializing in LLM post-training, optimization, agentic AI systems, and scalable ML infrastructure.

## TECHNICAL SKILLS

- Languages & Frameworks:** Python, PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph
- ML/AI Expertise:** LLMs (RAG, QLoRA, fine-tuning), VLMs, Vision Models (ViT, ConvNeXt, SAM2), Multimodal AI
- MLOps & Deployment:** Docker, Kubernetes, vLLM, AWS (SageMaker, Textract, Bedrock), Ray Serve
- Experimentation & Tools:** SQL, Pandas, NumPy, Elasticsearch, Airflow, Matplotlib

## WORK EXPERIENCE

### Principal Research Engineer, LLM Core & Agentic AI | Capital One (New York, NY) Jan 2026 – Present

- Developing multi-layer performance optimizations for agentic AI systems: expert-pruning on GPT-OSS to reduce model compute, latency/throughput benchmarking (TTFT, TPOT, tokens/sec) across batch sizes to inform model-selection decisions, and system-level optimizations within vLLM.
- Built a standardized multi-agent platform enabling downstream teams to productionize agentic workflows efficiently.
- Designed and implemented a pluggable memory system for agentic workflows, drawing on recent research (FluxMem, A-MEM) to enable dynamic, self-organizing memory that adapts its structure to interaction patterns rather than relying on fixed storage and retrieval.
- Implemented modular self-reflection loops using a generator-reflector-curator design (as in ACE), where a reflector distills lessons from execution feedback and a curator consolidates them into an evolving context playbook, improving agent performance under both offline and online context adaptation.

### Senior Machine Learning Engineer, R&D | CCC Intelligent Solutions (Chicago, IL) Jun 2022 – Dec 2025

- Trained policy-constrained LLMs using RLVR with GRPO, ranking group-sampled reasoning trajectories via automated insurance policy, consistency, and grounding verifiers to improve compliance and reasoning reliability.
- Led research and development of in-house LLM training and serving infrastructure, spanning dataset curation, QLoRA-based fine-tuning, evaluation harnesses, and multi-node deployments; optimized end-to-end throughput and latency using vLLM, accelerating experimentation and production readiness for large language models (LLaMA-4, Qwen).
- Developed a repair assistant agent using agentic RAG with automated retrieval evaluation, safety guardrails, and a voice interface, leveraging Whisper for speech-to-text, citation-based retrieval, and ReAct-style reasoning for grounded, auditable responses.
- Designed and trained ViT-based computer vision models to estimate chrome panel damage and other part-level vehicle attributes, incorporating custom loss functions to disambiguate visually similar operations

### Data Science & Cognitive Analyst | Deloitte (Bengaluru, India) Aug 2019 - Jul 2021

- Conducted rigorous EDA on medical datasets to identify key predictive features for patient outcomes, establishing performance baselines with POC models in AWS SageMaker to guide future development.

## EDUCATION

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|----------------------------------------------------------------------------------------------|--------------|
| <b>Columbia University</b>                                                                   | New York, NY |
| M.S. in Electrical Engineering, Specialization in Machine Learning, CGPA: 4.09/4             | Dec 2022     |
| Coursework: Machine Learning, Deep Learning, Computer Vision, Algorithms, Big Data Analytics |              |

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|-----------------------------------------------------------------------------|----------------|
| <b>Manipal Academy of Higher Education, Manipal Institute of Technology</b> | Manipal, India |
| B.Tech. in Electrical and Electronics Engineering, CGPA: 8.35/10            | May 2019       |

## AWARDS

- Recipient of the MS EE Award of Excellence, a distinction for top achievement granted annually to less than 5% of all MS students.

## PUBLICATIONS

- Real-time Multi-Camera Analytics for Traffic Information Extraction and Visualization** – 2023 IEEE International Conference on Pervasive Computing, March 2023